

JIAWEI LIU

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EDUCATION

Carnegie Mellon University

Aug 2023 – May 2027 (Expected)

BESA in Electrical & Computer Engineering and Music Technology

Pittsburgh, PA

- Coursework: Digital Signal Processing, Electroacoustics, Intro to Computer Music, Deep Learning, Generative AI

Dwight School Seoul

Aug 2020 – May 2023

International Baccalaureate Diploma

Seoul, South Korea

- Higher Level courses: Physics, Psychology, Music
- Research papers on string behavior in instruments ([link](#)), music theory analysis ([link](#)), and golf simulation modeling ([link](#))

EXPERIENCE

Research Intern

Sep 2026 – Present

G-Clef (Generative Creativity Lab), Carnegie Mellon University — advised by Prof. Chris Donahue

Pittsburgh, PA

- Collaborating with Yewon Kim on an ongoing research project

Research Intern, Course Development

May 2026 – Aug 2026

Intro to Computer Music (15-322/622), Carnegie Mellon University — advised by Prof. Chris Donahue

Pittsburgh, PA

- Co-developed the Fall 2026 course redesign, building its interactive online textbook and website with Jupyter Book and MyST
- Added live in-browser Python (Pyodide), interactive Plotly widgets, and Manim animations to teach DSP concepts
- Built and published **browseraudio** (PyPI), an open-source library for recording and playing audio in browser-based Python
- Wrote new assignments on convolution, Fourier analysis, filters, and reverb, with Python autograder tests for Gradescope

Teaching Assistant

Jan 2026 – Present

Intro to Computer Music (15-322/622), Carnegie Mellon University — taught by Prof. Chris Donahue

Pittsburgh, PA

- Support a programming- and project-based course on digital audio, sound synthesis, and audio signal processing
- Hold office hours, answer student questions on Piazza, lead review sessions, and grade assignments and final projects

Independent Researcher

Jun 2022 – Dec 2022

Music Generation with RNNs ([link](#)) — advised by Dr. Yong Zhuang, Research Scientist at Bloomberg AI

Seoul, South Korea

- Trained an LSTM-based recurrent neural network on a large dataset of musical notation to generate music
- Model predicts each subsequent note from the preceding sequence, showing deep learning's potential in creative domains

PROJECTS

Pedal Morphing Instrument | BXA Capstone — advised by Prof. Chris Donahue | [Project Page](#)

Aug 2026 – Present

- Building a guitar-effect instrument that morphs continuously between real pedals so guitarists can design new hybrid tones
- Designing a VAE to learn a continuous tone space from recordings of 10–20 overdrive, distortion, and fuzz pedals
- Planning a real-time plugin with a 2D tone plane and text control, to be evaluated in recorded studio sessions with guitarists

Piano Genie Max Edition | Twisted Signals Final Project, Carnegie Mellon University

Apr 2026

- Rebuilt Piano Genie in Max 9, running the Magenta.js model through Node for Max so eight computer keys play a full piano
- Wrote a JavaScript style arranger that re-voices played notes in real time as classical, jazz, or J-pop in any key and mode
- Built an 8-voice sampler (piano, violin, organ, guitar) with custom chorus, stereo delay, and reverb effects

Intergram | [Colab](#), [Presentation](#)

Mar 2024

- Led a four-person team at the Google-sponsored GenAI Hackathon, finishing in the top 4 of 26 teams
- As primary developer, built a Python demo of a multilingual messaging app where each user chats in their own language

CMUapartment.com | [Site](#), [GitHub](#)

Jan 2024

- Led a three-person team at a 24-hour hackathon to build a CMU housing site that ranks apartments by each student's priorities

SKILLS

Programming: Python (NumPy, pyquist), C, MATLAB, JavaScript, HTML/CSS, Git, GitHub Actions, Jupyter Book, Pyodide

Audio & Music: Audio DSP, sound synthesis, Web Audio API, Max/MSP, Logic Pro, Ableton Live, Pro Tools, mixing & mastering

Languages: English & Chinese (native), Korean (daily conversational)

LEADERSHIP & ACTIVITIES

Audio Engineering Society (AES): Founding member, Carnegie Mellon University student section

Badminton: President and competitive player, Carnegie Mellon badminton team